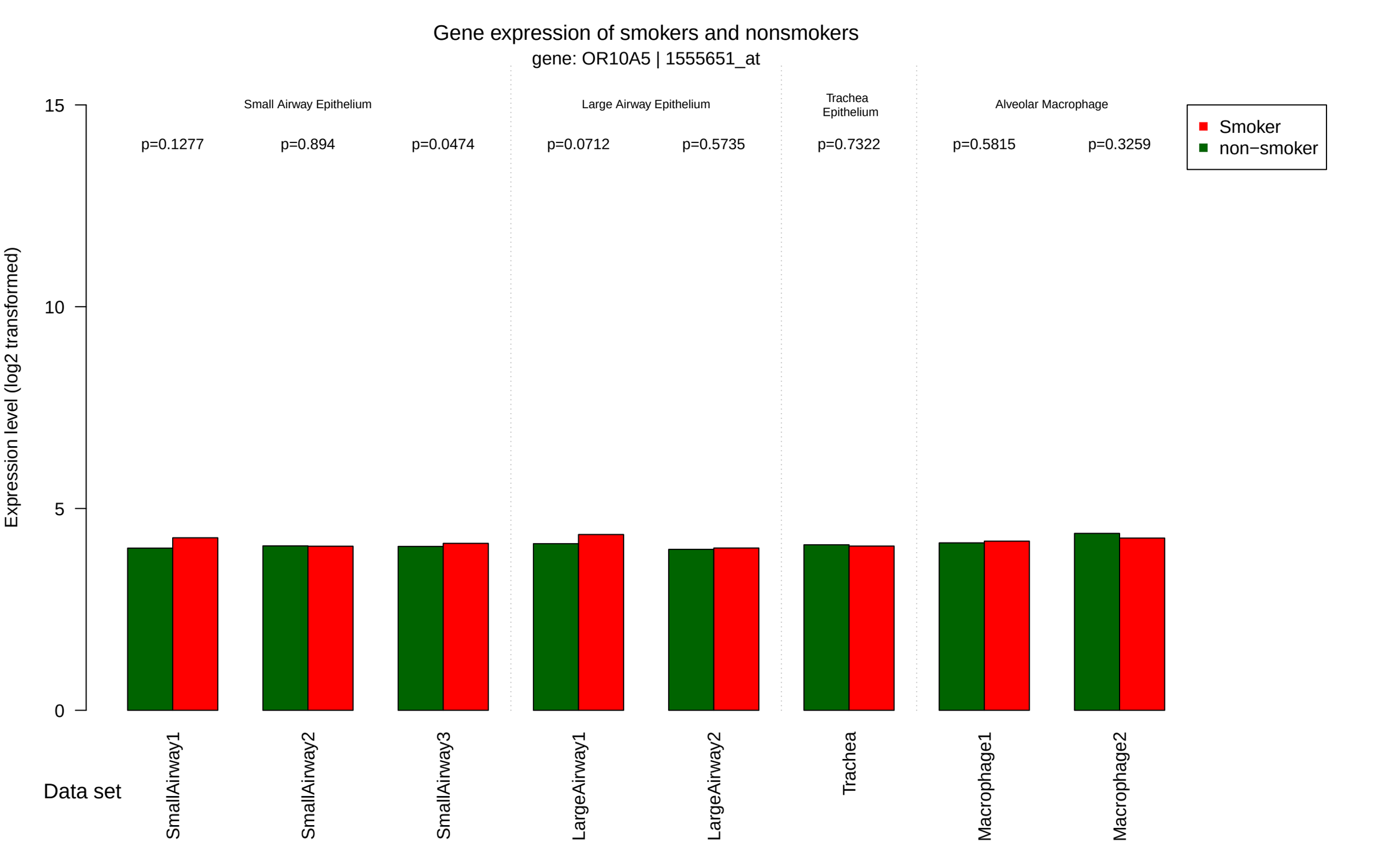
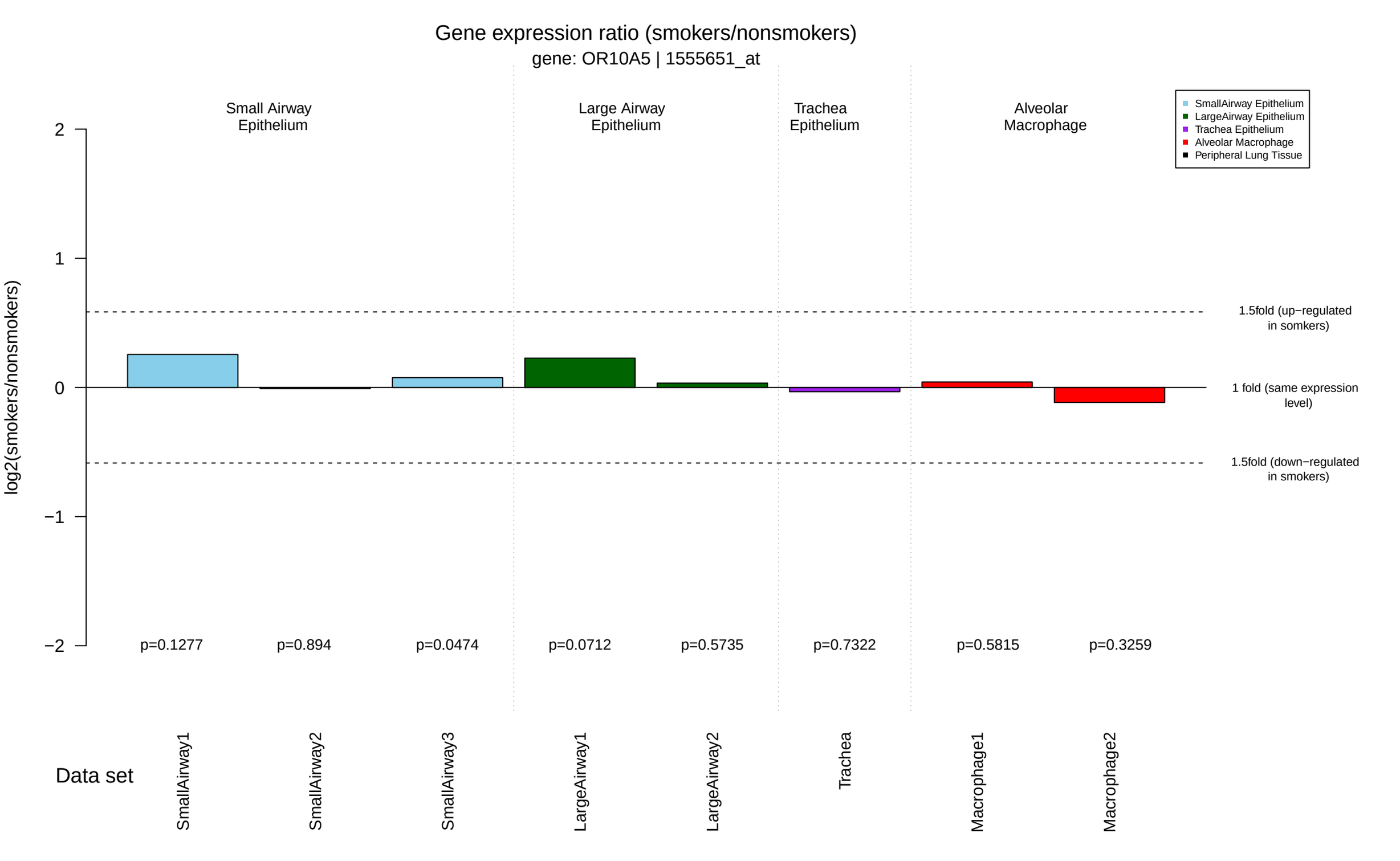
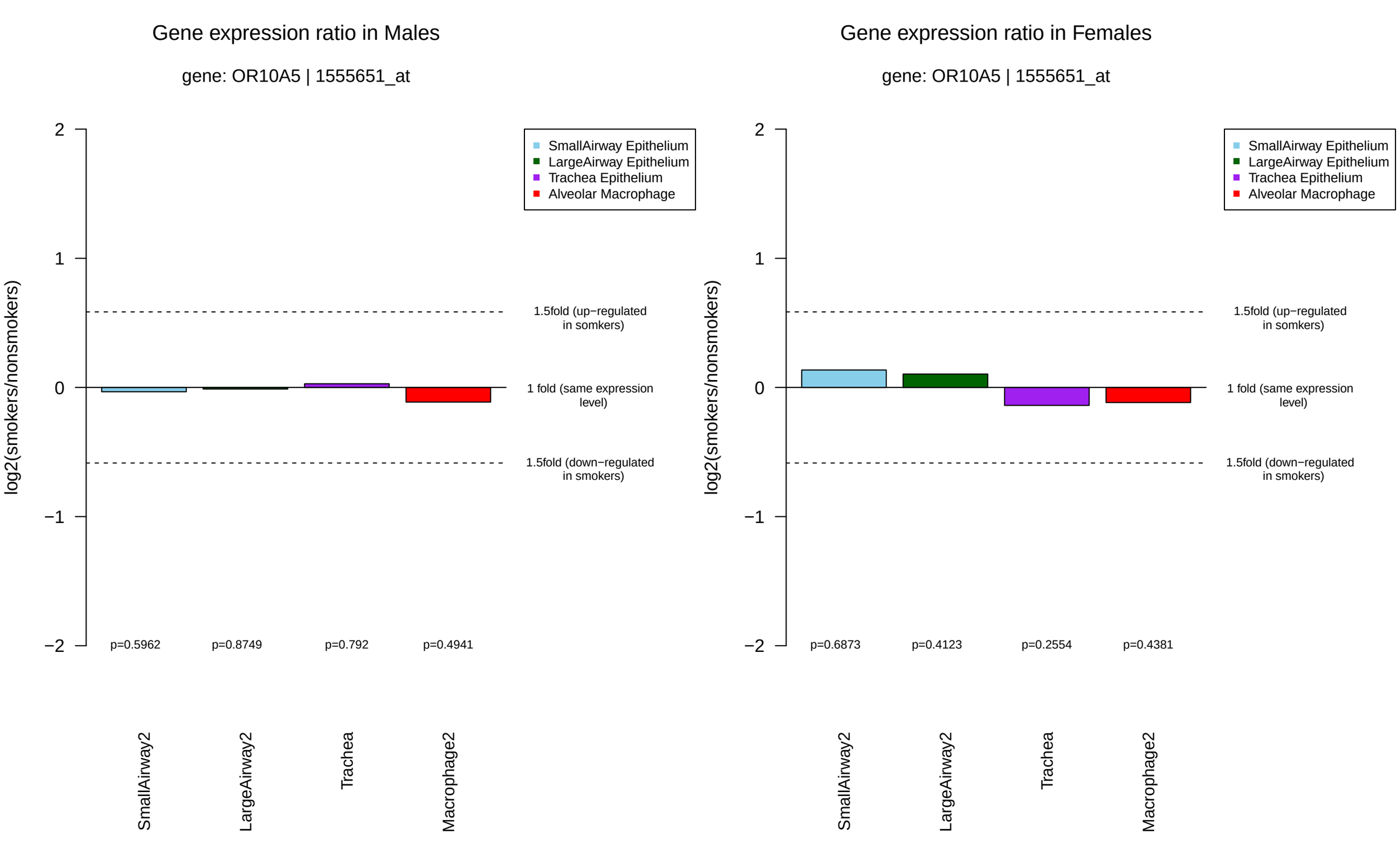
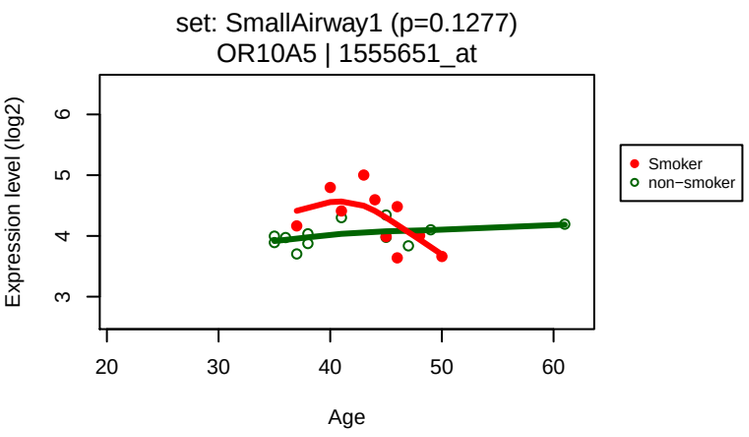
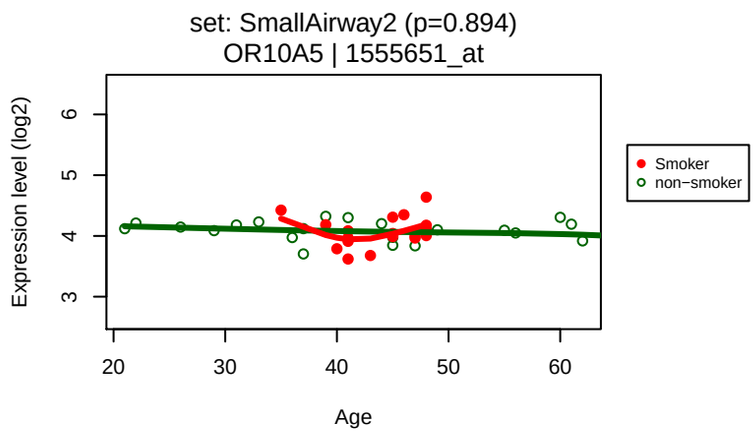
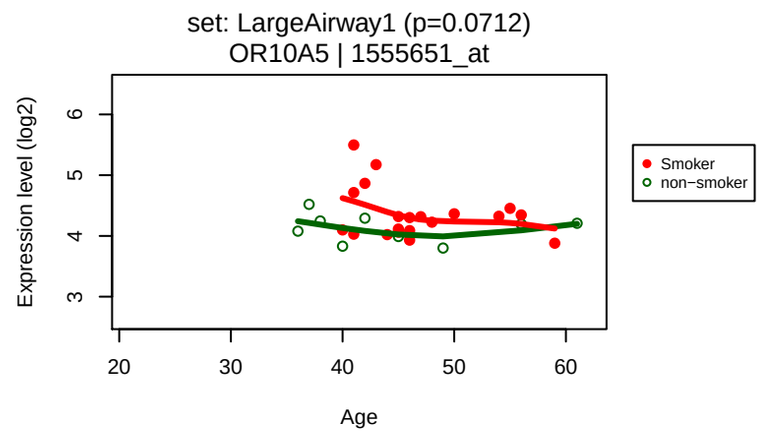

Gene expression of smokers and nonsmokers
gene: OR10A5 | 1555651_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1277
p=0.894
p=0.0474
p=0.0712
p=0.5735
p=0.7322
p=0.5815
p=0.3259
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

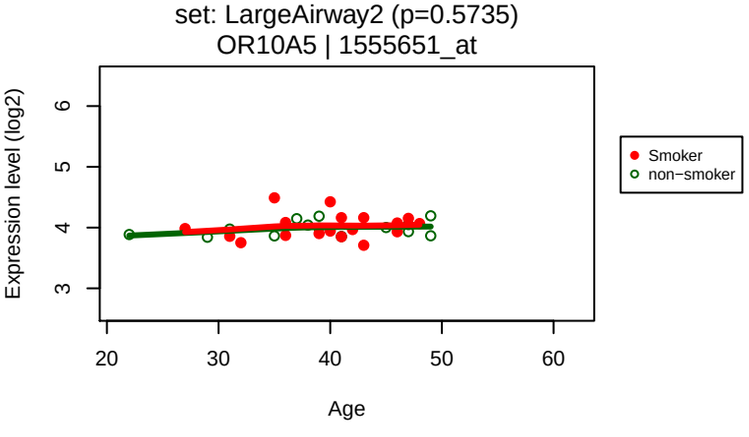

Gene expression ratio (smokers/nonsmokers)
gene: OR10A5 | 1555651_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1277
p=0.894
p=0.0474
p=0.0712
p=0.5735
p=0.7322
p=0.5815
p=0.3259
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

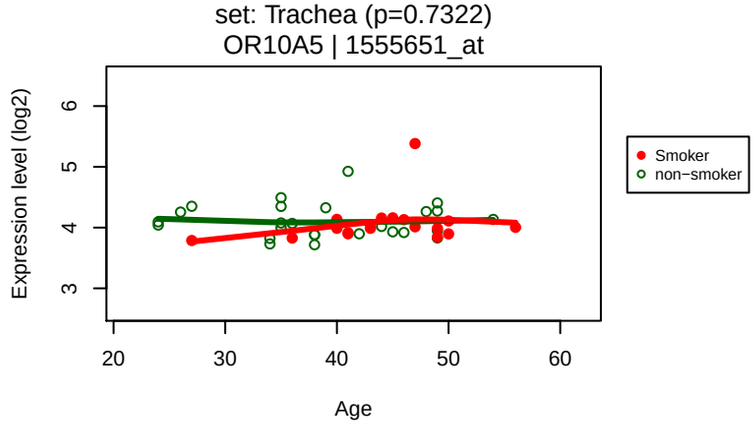

Gene expression ratio in Males
gene: OR10A5 | 1555651_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5962
p=0.8749
p=0.792
p=0.4941
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: OR10A5 | 1555651_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6873
p=0.4123
p=0.2554
p=0.4381
SmallAirway2
LargeAirway2
Trachea
Macrophage2

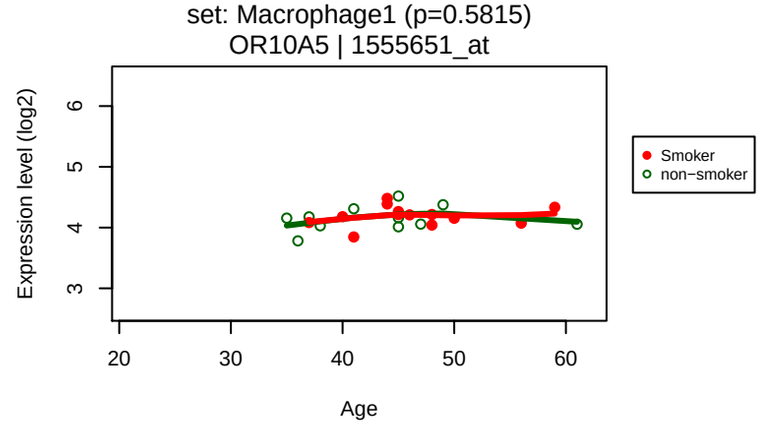

set: SmallAirway1 (p=0.1277)
OR10A5 | 1555651_at
Expression level (log2)
Age
Smoker
non−smoker

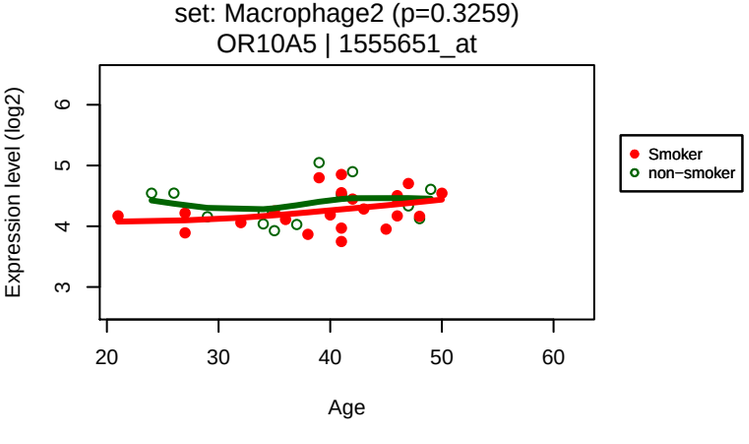

set: SmallAirway2 (p=0.894)
OR10A5 | 1555651_at
Expression level (log2)
Age
Smoker
non−smoker

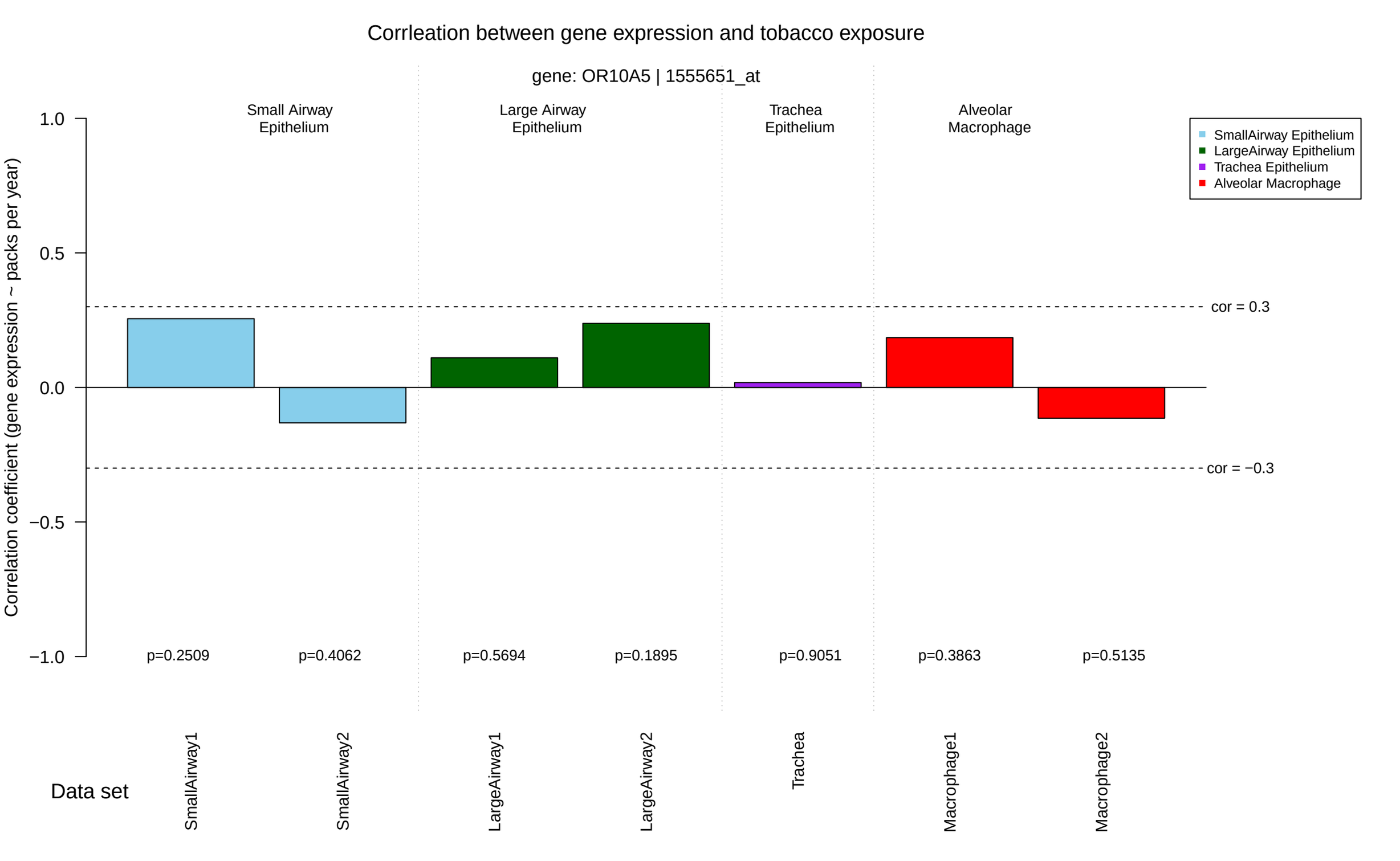

set: LargeAirway1 (p=0.0712)
OR10A5 | 1555651_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5735)
OR10A5 | 1555651_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7322)
OR10A5 | 1555651_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5815)
OR10A5 | 1555651_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3259)
OR10A5 | 1555651_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: OR10A5 | 1555651_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2509
p=0.4062
p=0.5694
p=0.1895
p=0.9051
p=0.3863
p=0.5135
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2